

Practice Problem Set 5

ECON 101 – Macroeconomics

Winter 2026

1. Explain the three functions of money:

- medium of exchange,
- unit of account,
- store of value.

Provide one real-world example for each.

2. Distinguish between:

- M1 and broader measures of money supply,
- currency vs. deposits.

Why are some assets included in the money supply while others are not?

3. Explain how commercial banks create money. What role does the reserve ratio play in this process?

4. Deposit Creation

Suppose:

- Initial deposit = \$5,000,
 - Desired reserve ratio = 0.10.
- (a) Calculate required reserves.
- (b) Calculate initial loans.
- (c) Calculate the simple deposit multiplier.
- (d) Calculate the maximum increase in deposits.

5. Leakages in the Banking System

Explain how the following reduce the money multiplier:

- excess reserves,
- currency held by the public.

6. Explain how a bank run occurs and how the Bank of Canada and deposit insurance help prevent or limit bank runs.

7. What is the Bank of Canada's inflation target? Why is inflation targeting important?
8. Explain the monetary transmission mechanism step-by-step starting from a change in the policy interest rate.
9. Distinguish between:
 - expansionary monetary policy,
 - contractionary monetary policy.

When would each be used?

10. Multiplier and Monetary Policy

Suppose:

- $MPC = 0.8$,
 - Investment increases by \$40 billion due to lower interest rates.
- (a) Calculate the multiplier.
 - (b) Calculate the total change in GDP.

11. Interest Rate Effects

Explain how an increase in interest rates affects:

- consumption,
 - investment,
 - exchange rate,
 - net exports.
12. Why might monetary policy be less effective during a recession? Discuss at least two reasons.

13. Distinguish between:

- discretionary fiscal policy,
- automatic stabilizers.

Provide examples of each.

14. Explain how government spending and taxes affect aggregate demand.
15. What is a government budget deficit and surplus? How are they related to the business cycle?

16. Government Spending Multiplier

Suppose:

- $MPC = 0.75$,
- Government spending increases by \$80 billion.

- Calculate the multiplier.
- Calculate the change in GDP.

17. Tax Multiplier

Suppose:

- $MPC = 0.75$,
- Taxes decrease by \$60 billion.

- Calculate the tax multiplier.
- Calculate the change in GDP.

18. The economy is in a recession:

- GDP is below potential,
- unemployment is rising.

- Use the AE model to explain the recession.
- Use the AD–AS model to illustrate the situation.
- Propose one fiscal policy and one monetary policy. Explain how each affects GDP.

19. Suppose government increases spending by \$100 billion but finances it through borrowing.

Explain:

- how interest rates change,
- how private investment is affected,
- why the actual increase in GDP may be smaller than predicted.

End of Problem Set